Simulation-Driven Design and Functional Assessment of a Gait Simulator for Transtibial Prosthesis Evaluation

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***Abstract*—Biomechanical gait simulators have become promising alternatives for prosthesis evaluation under controlled laboratory conditions. This work presents the design, implementation, and functional assessment of a three-degree-of-freedom (3-DOF) gait simulator for evaluating transtibial prostheses, featuring horizontal, vertical, and sagittal motion axes. The proposed platform adopts a reduced-degree-of-freedom architecture developed following a simulation-driven design methodology, combining CAD modeling, finite element analysis, analytical sizing of critical transmission components, and the integration of custom mechanical, electrical, electronic, and embedded control hardware. A functional assessment was performed using gait trajectories acquired from a healthy subject, marker-based video motion analysis, and force plate measurements. The proposed device reproduced the tibial-segment inclination angle with $\mathrm{RMSE}_{\mathrm{norm}}$ values of 0.38 and 1.58 during the stance and swing phases, respectively, with correlation coefficients of 1.00 and 0.997 and intra-device inter-trial ICC(3,1) values above 0.999. For the vertical ground reaction force during the stance phase, an $\mathrm{RMSE}_{\mathrm{norm}}$ of 21.87, a correlation coefficient of 0.9501, and an ICC(3,1) of 0.9984 were obtained. These results indicate that the proposed platform can accurately and repeatably reproduce gait kinematics, providing preliminary evidence that a reduced-degree-of-freedom simulation-driven architecture can serve as a practical experimental platform for evaluating lower-limb prostheses.**

***Index Terms*—Gait simulator, Ground reaction force, Marker-based motion analysis, Mechanical design, Transtibial prosthesis**

I. Introduction

Human gait is a complex biomechanical process that relies on coordinated interactions among joints, muscles, and ground reaction forces to maintain balance and stability, absorb impact, and generate forward propulsion [1]–[3]. In individuals with lower-limb amputation, these interactions are altered by the prosthetic device. Although a degree of gait asymmetry reflects natural functional differences between the lower limbs in able-bodied walking [4], amputee gait is characterized by asymmetries amplified beyond that range, together with increased energy expenditure and abnormal loading patterns that may compromise mobility and long-term musculoskeletal health [5], [6]. Consequently, the development and evaluation of lower-limb prostheses remain important challenges in biomechanics, rehabilitation engineering, and prosthetic design.

Conventional prosthetic evaluation relies on mechanical testing, clinical gait analysis, and experiments involving human participants. Although these methods provide valuable clinical information, they are often expensive, time-consuming, subject to inter-subject variability, and may pose safety risks during the evaluation of early-stage prototypes or active prosthetic systems. These limitations have motivated the development of biomechanical gait simulators capable of reproducing controlled, repeatable walking conditions for preclinical evaluation of assistive devices [7], [8]. Existing approaches include mechanical test benches, robotic gait simulators, musculoskeletal simulations, hardware-in-the-loop platforms, and prosthesis simulators worn by participants without lower-limb amputation. However, each approach presents specific limitations. Hardware-in-the-loop platforms require real-time coupling between the physical prosthesis and a numerical model [9]. Robotic gait simulators reproduce multi-planar motion, but at a capital cost that constrains their accessibility [10], [11].Simulation-based approaches avoid hardware constraints altogether but do not evaluate physical prototypes [12]–[15].

Prosthesis simulators worn by participants without amputation introduce confounding factors that are difficult to separate from the device under test: leg-length discrepancies above 2 cm alter gait kinematics, the folded sound limb introduces an inertial artifact that can exacerbate step-length asymmetry, and the added mass increases metabolic cost [7]. Despite these constraints, such platforms enable the systematic assessment of prosthetic alignment, stiffness, damping, and control strategies while reducing reliance on early-stage human testing [9], [10], [16]–[20].

Despite this potential, recent literature reviews indicate that most existing lower-limb prosthesis simulators have focused on transfemoral applications, with transtibial-specific platforms comparatively underrepresented [7]. In addition, the development of gait simulators has been shown to involve an inherent trade-off between the number of controlled degrees of freedom and system cost and complexity, with high-DOF platforms requiring substantial capital investment and reduced-DOF designs facing a corresponding compromise in simulation accuracy [21]. These observations motivate the development of a mechanically simplified, transtibial-oriented gait simulator capable of reproducing the essential kinematic and kinetic demands of gait within a reduced-degree-of-freedom mechanical architecture.

This work presents the design, implementation, and functional assessment of a 3-degree-of-freedom (3-DOF) gait simulator for transtibial prosthesis testing, providing two translational axes (horizontal and vertical) and one rotational axis (sagittal). The architecture was developed using a simulation-driven design methodology that integrated finite element analysis, analytical transmission sizing, and functional verification. This study describes the design and implementation processes, together with a functional assessment using videogrammetry and ground reaction force (GRF) analysis.

II. System Architecture

A. Mechanical Design

The gait simulator was developed using a *Simulation-Driven Design* approach that combined CAD modeling, standardized industrial components, analytical transmission sizing, and finite element analysis before manufacturing. The final architecture comprises three independently actuated coordinates: horizontal translation, vertical translation, and sagittal-plane rotation, selected to reproduce the position and orientation of the prosthetic test assembly while maintaining mechanical simplicity. The design also incorporated six limit switches for travel protection and homing, compatibility with the force platform and marker-based motion analysis, and a rigid interface for mounting transtibial prostheses.

1) Mechanical Architecture: The mechanical assembly was modeled in Autodesk Inventor and implemented as a modular welded-steel frame with an approximate footprint of 1153×974 mm and a total height of 1424 mm. Structural plate thicknesses ranged from 3 to 12 mm. Fig. 1 shows the arrangement of the three motion modules and the principal transmission elements.

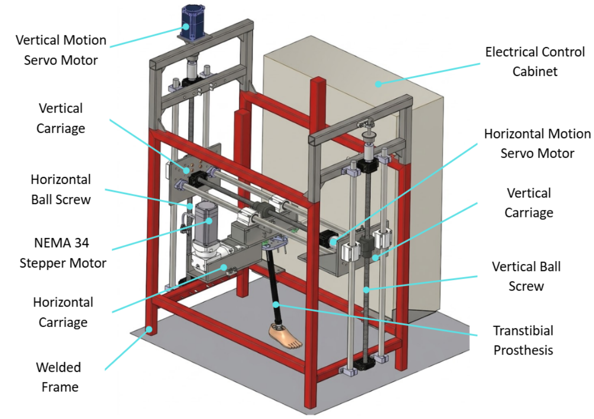


Fig. 1. CAD model of the final mechanical architecture and principal motion modules, including the transtibial prosthesis mounted on the platform

The horizontal axis provides approximately 587 mm of travel using an SFU2510 ball screw, WCS25 shafts, SC25UU linear bearing blocks, and an A6M80-750H2A1-M17 servomotor. The vertical axis provides approximately 542 mm of travel using SFU2505 ball screws, linear guides, a roller-chain transmission, and an A6M80-750H2B1-M17 servomotor with a holding brake. Sagittal rotation ranges from $-44°$ to $47°$ and is generated by a 34E1K-120 closed-loop stepper motor coupled to an RYG34-G50 planetary gearbox with a 50:1 reduction ratio. The three axes converge at a 150×120 mm prosthetic mounting platform positioned approximately 752 mm above the ground reference in the nominal configuration.

2) Structural Verification and Transmission Selection: Static finite element analyses were performed in ANSYS 2025 R2 for the main frame, horizontal carriage, left vertical support plate, right vertical carriage, and electrical-cabinet support. The simulations guided local reinforcement and load-transfer improvements before manufacturing. Table I summarizes the maximum deformation, equivalent von Mises stress, and static factor of safety obtained for each assembly, assuming a structural-steel yield strength of approximately 250 MPa.

TABLE I
Summary of finite element analysis results for the critical mechanical assemblies.

Assembly	Max. def. (mm)	Max. stress (MPa)	FoS
Main frame	0.0659	36.219	6.9
Horizontal carriage	0.027	18.226	13.7
Left vertical support plate	0.215	198.060	1.3
Right vertical carriage	0.069	39.642	6.3
Electrical-cabinet support	0.267	29.094	8.6

The left vertical support plate was the critical component, reaching a maximum equivalent stress of 198.06 MPa, a maximum deformation of 0.215 mm, and a minimum static factor of safety of approximately 1.3. All analyzed assemblies remained below the assumed yield strength under the considered static load cases. These results supported the release of the refined design for manufacturing; fatigue life and long-term cyclic durability were not inferred from this analysis.

For the vertical-axis chain transmission, a corrected design

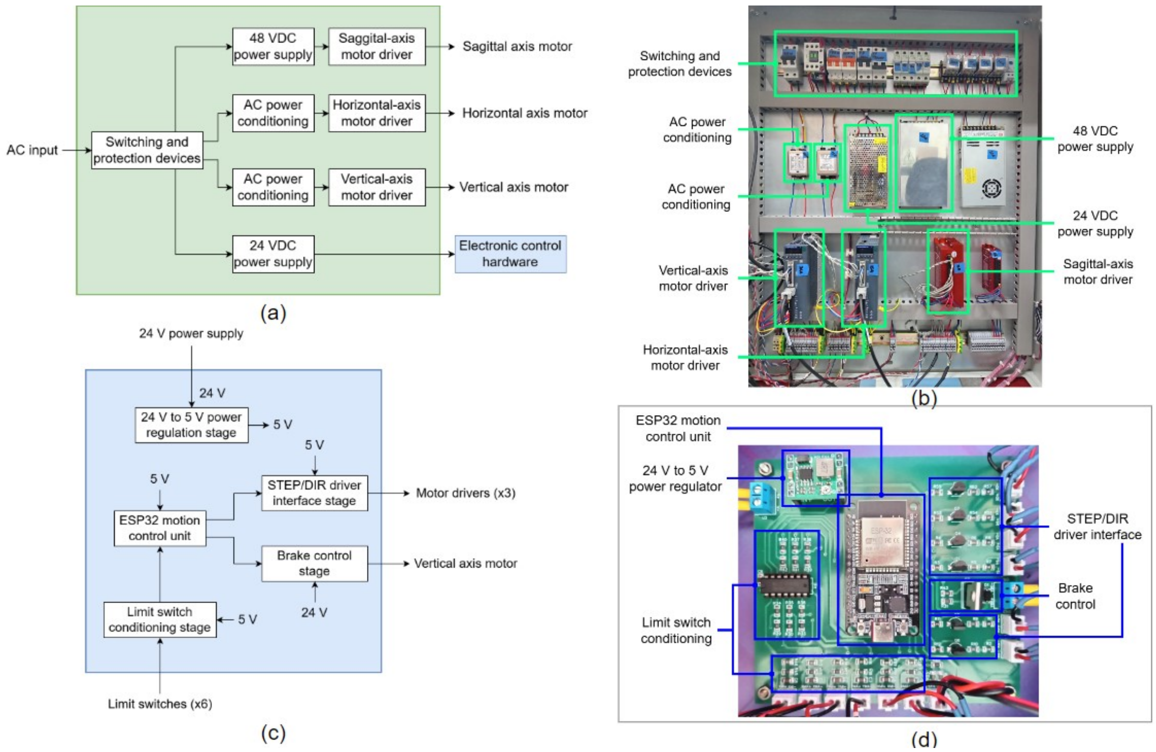


Fig. 2. Electrical power system and electronic control hardware: (a) block diagram of the electrical power system; (b) implemented electrical panel; (c) block diagram of the electronic control hardware; (d) fabricated electronic control board.

power of $1.05\ \mathrm{kW}$ was obtained from the $0.75\ \mathrm{kW}$, $3000\ \mathrm{rpm}$ servomotor using a service factor of 1.4 and a 19-tooth driving sprocket. An ANSI 35 single-strand roller chain was selected as a compromise among load capacity, installation space, availability, and cost. The final transmission uses ASA 35B19 sprockets and an ASA 35-1R chain with 244 pitches, corresponding to a nominal length of $2324.1\ \mathrm{mm}$.

B. Electrical Power System

Fig. 2(a) presents a block diagram of the power system, which supplies the motor drivers and the electronic control hardware.

Single-phase AC mains power first passes through a switching and protection stage composed of rotary cam switches, thermomagnetic circuit breakers, magnetic contactors, fuses, and relays. The horizontal- and vertical-axis motor drivers are supplied from the AC mains through dedicated line filters. A 48 VDC power supply powers the sagittal-axis motor driver. In addition, a 24 VDC power supply powers the electronic control hardware and auxiliary external devices.

The horizontal- and vertical-axis motors are driven by A6-750RS closed-loop AC servo drives, which convert single-phase AC input into three-phase synchronous output for motor operation. The sagittal-axis motor is controlled by a CL86Y closed-loop stepper driver operating from a 48 VDC power supply. Fig. 2(b) shows the implemented electrical panel.

C. Electronic Control Hardware

The electronic hardware was designed around an ESP32 development board. This hardware provides the interfaces required to interact with the motor drivers, the vertical-axis motor brake, and the limit switches. Fig. 2(c) shows a block diagram of this concept. An MP1584EN regulator module steps down the 24 VDC supplied by the electrical power system to 5 VDC. An ESP32 module serves as the central processing unit, executing the embedded software that generates pulse and direction signals for the motor drivers, controls the vertical-axis motor brake, manages the homing routine through the limit switches, and supervises overall system operation.

Since the motor drivers employ opto-isolated pulse and direction inputs, transistor-based interface circuits were designed to provide the required drive current and ensure reliable signal transmission at pulse frequencies of up to 100 kHz.

Since the vertical-axis motor incorporates a built-in electromagnetic brake to prevent unintended displacement when the system is powered down, a dedicated switching circuit was implemented to control the brake release and engagement during operation. Finally, RC filtering and Schmitt-trigger conditioning were applied to the limit-switch signals for hardware debouncing and noise immunity during homing and normal operation. Fig. 2(d) shows the fabricated electronic board indicating its functional blocks.

D. Motion Control Software

The motion control software consists of an offline trajectory processing stage and an embedded motion control stage. In the first stage, experimental gait data recorded in CSV format are processed using a Python script that converts the trajectories into a C++ header file containing the trajectory

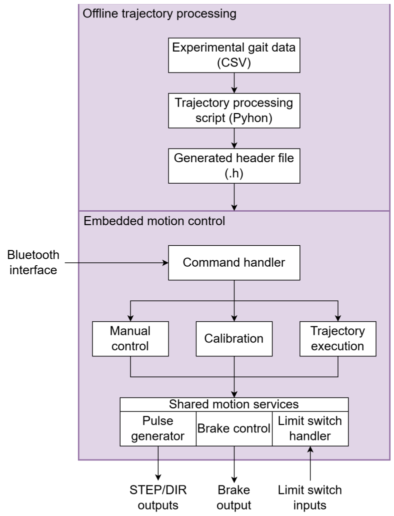


Fig. 3. Block diagram of the motion control software.

arrays and homing parameters, which are then compiled into the embedded firmware.

The embedded motion control stage is implemented on an ESP32 development board. This stage manages user interaction, coordinates three-axis motion, and ensures safe execution of the gait trajectories. Fig. 3 shows the software architecture comprising both stages.

User commands are received via a Bluetooth interface and processed by a command handler that supports three operating modes: manual control (direct incremental motion of each axis), calibration (axis homing via limit switch detection to establish reference positions), and trajectory execution (synchronized three-axis motion using the stored trajectory arrays).

The shared motion services include a pulse generation module that generates STEP/DIR signals for all motor drivers, depending on the selected operating mode, without blocking execution. All three operating modes share this module. Safe operation is ensured by a limit-switch-handling interrupt routine that protects against motion beyond the physical travel limits. Additionally, the system controls the vertical-axis motor brake during initialization and operation.

III. Functional Assessment

A functional assessment was conducted to verify the capability of the simulator's movement relative to that of a reference subject (male, 86 kg, 1.74 m). The same instrumentation and setup were used to acquire the gait trajectories that programmed the simulator and to evaluate its output, ensuring methodological consistency.

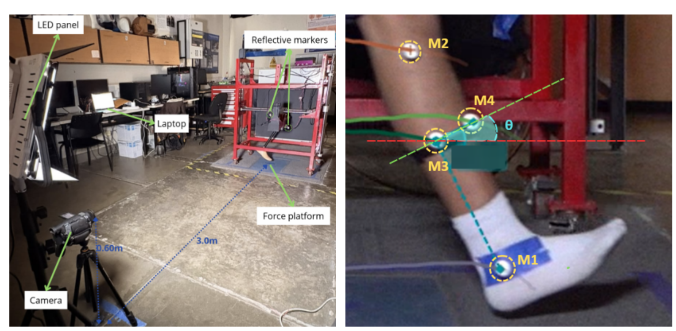
Kinematic and kinetic assessments were conducted in independent experimental sessions, corresponding to separate validation conditions (without and with a mounted prosthesis, respectively); therefore, temporal synchronization between the camera and the force platform was not required.

Kinematic data were recorded using a Sony FDR-AX700 camera at 120 fps, positioned perpendicular to the simulator's sagittal plane at a distance of 3 m and a height of 60 cm above ground level, as shown in Fig. 4(a). Reflective markers measuring 2.5 cm in diameter were used to identify reference points and were illuminated by a Godox Litemons LP1200R LED panel (120 W, 100% intensity, white light mode) to ensure consistent marker visibility.

Four reflective markers (M1–M4) were placed on the reference subject: M1 at the lateral malleolus and M2 42 cm proximal to it along the transtibial segment, with its position used to calculate the horizontal and vertical trajectories. M3 was placed at the midpoint of the segment formed by M1 and M2, and M4 was aligned with M3 to form a segment perpendicular to the tibial segment, directed anteriorly. On the simulator, two reflective markers were placed at the ends of the moving platform to track its trajectory under the same protocol. The tibial-segment inclination angle θ was defined as the orientation of the segment perpendicular to the tibial axis (M3–M4) relative to the horizontal image axis, $\theta = \operatorname{atan2}(y_{M4} - y_{M3}, x_{M4} - x_{M3})$, positive above the horizontal and negative below; equivalently, the deviation of the tibial segment from the vertical. The same definition was applied to the two simulator markers spanning the moving platform, which is perpendicular to the prosthesis axis. Marker placement is shown in Fig. 4(b).

Camera calibration for pixel-to-metric conversion was performed in Kinovea using the width of the AMTI force platform (40 cm), located within the same motion plane, as the reference object to minimize angular distortion errors. Data were processed using Kinovea v.2025.2 and MATLAB R2025b. Ground reaction forces were acquired using an AMTI BP400600 force platform at 1000 Hz. Both the reference and simulator GRF signals were normalized to the subject's body weight (86 kg 843.7 N) and expressed as %BW.

Marker coordinates were smoothed with a third-order Savitzky–Golay filter over a 9-frame window, and the vertical force with a zero-phase fourth-order Butterworth low-pass filter at 15 Hz. Initial contact and toe-off were detected on the filtered force using a 20 N threshold [22]. Stance and swing were time-normalized to 0–60% and 60–100% of the gait cycle and resampled by piecewise cubic interpolation. Marker occlusions were resolved by manual re-tracking in Kinovea.



(a) (b)

Fig. 4. (a) Experimental setup for kinematic and kinetic data acquisition. (b) Placement of markers M1–M4 on the reference subject and definition of the tibial-segment inclination angle θ.

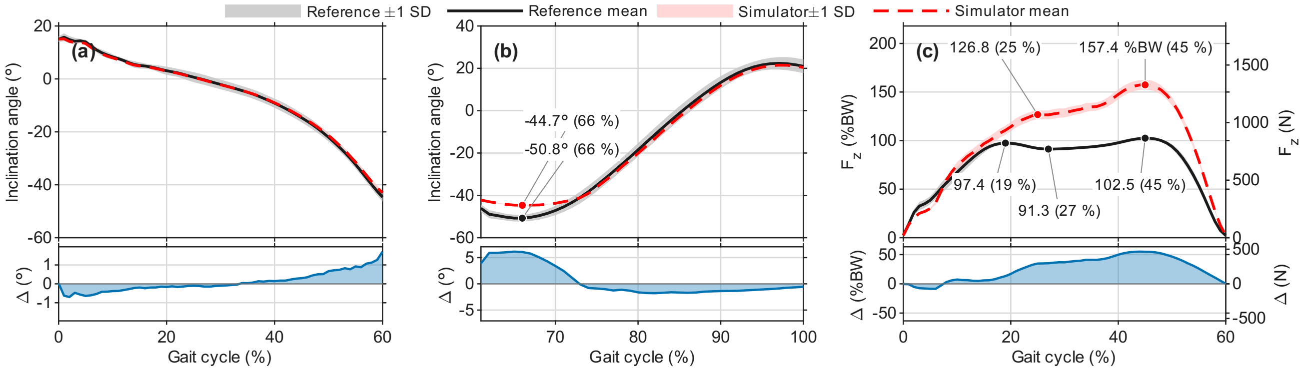


Fig. 5. Functional assessment of the gait simulator. Tibial-segment inclination angle during (a) stance and (b) swing, and (c) vertical ground reaction force during stance. Markers label each extremum, with its instant of occurrence in parentheses as a percentage of the gait cycle. Δ is the pointwise residual, simulator − reference.

The trajectory data were acquired from the reference subject, whose gait cycles were recorded for the stance and swing phases. The captured trajectories were then exported as CSV files containing time, horizontal and vertical position coordinates, and the tibial segment inclination angle. For each phase, ten gait cycles were recorded and averaged to obtain the reference trajectory used to program the simulator. To characterize the intra-subject variability associated with this averaging process, the RMSE between the individual cycles and the resulting mean curve was calculated for the tibial inclination angle, yielding values of 1.41° during stance and 2.53° during swing, thereby confirming the acceptable consistency of the reference data.Stance and swing were captured separately, lasting 0.95 s and 0.55 s in the subject and 28.8 s and 15.9 s in the simulator. The subject's mean walking speed, computed from the horizontal displacement of the tibial segment over each phase, was 0.48 m/s in stance and 0.97 m/s in swing.

Ten repetitions of the simulator's programmed trajectory were recorded for each phase to assess output repeatability, independently of the ten gait cycles used to derive the reference trajectory.

Two validation conditions were evaluated: (1) kinematic fidelity without a prosthesis, comparing the tibial-segment inclination angle during the stance and swing phases independently; and (2) kinetic fidelity with a passive commercial transtibial prosthesis (42 cm prosthetic shank length, articulated prosthetic foot) mounted on the simulator, comparing the vertical ground reaction force (Fz, %BW) during the stance phase only, since no ground contact occurs during the swing phase.The simulator's tracking error with respect to the reference trajectory was quantified using RMSE normalized by the pointwise reference standard deviation ($\mathrm{RMSE}_{\mathrm{norm}}$),waveform similarity using the Pearson correlation coefficient (r), and the percentage of simulator data points within ±1 SD of the reference, which expresses the same normalization point by point.In addition, the intra-device inter-trial repeatability of the simulator across its ten programmed repetitions was quantified independently using the intraclass correlation coefficient ICC(3,1).

The simulator reproduced the tibial inclination angle with $\mathrm{RMSE}_{\mathrm{norm}}$ = 0.38, r = 1.00, and 100% of points within ±1 SD during the stance phase. The ICC(3,1) was 0.999, confirming high repeatability across the ten simulator trials (Fig. 5(a)). During the swing phase, the corresponding values were $\mathrm{RMSE}_{\mathrm{norm}}$ = 1.58, r = 0.997,and 72.50% of points within ±1 SD, with an ICC(3,1) of 0.999 (Fig. 5(b)). For the vertical ground reaction force, $\mathrm{RMSE}_{\mathrm{norm}}$ = 21.87 and r = 0.9501 were obtained during the stance phase,with an ICC(3,1) of 0.9984, confirming consistent force output across the ten simulator repetitions(Fig. 5(c)).

IV. Discussion

The functional assessment confirmed that the simulator accurately reproduces the reference subject's tibial-segment inclination angle during stance and swing,with ICC(3,1) values above 0.999 in both phases, confirming high repeatability of the simulator's motion across trials. A deviation was observed between approximately 60% and 75% of the gait cycle, where the simulator angle stabilizes near its mechanical limit while the reference subject continues towards more negative inclination values. This deviation is not attributable to a loss of fidelity, as the simulator trajectory remains aligned with the reference trajectory throughout the swing phase, but rather to the mechanical range of the sagittal rotation axis (-44° to 47°), which restricts the achievable rotation during early swing. This observation constitutes a concrete, quantifiable design constraint that can directly inform future revisions of the mechanical architecture.

For the vertical ground reaction force during the stance phase, a strong waveform correlation was obtained (r = 0.9501, ICC(3,1) = 0.9984), confirming consistent force output and accurate reproduction of the characteristic double-peak pattern of normal gait. However, the magnitude of the peak force exceeded the reference values. This overestimation can be qualitatively attributed to the added structural mass of the simulator's moving assembly and to the passive mechanical behavior of the commercial prosthesis used in this condition, rather than to inaccuracies in tracking the horizontal and vertical trajectories. Manufacturer specifications for the prosthesis, including stiffness properties, were unavailable. Similarly, the added mass introduced by the simulator's moving assembly was not quantified in this study. Direct measurement of both parameters is recommended in future work to isolate their respective contributions to the observed force overestimation.

Overall, these findings indicate that the mechanical design and control architecture of the simulator can reproduce the fundamental kinematic and kinetic patterns of human gait with high fidelity and repeatability.The kinematic fidelity reported above was obtained with an architecture simplified with respect to multi-degree-of-freedom robotic simulators: frontal- and transverse-plane motion is not actuated, while the three sagittal-plane coordinates governing the assessed variables are retained. These same three coordinates, flexion–extension with vertical and horizontal translation, have been identified as those required for prosthetic testing [21], in that case applied to prosthetic knees rather than to transtibial prostheses.

V. Conclusion

This paper presents the design, implementation, and functional assessment of a three-degree-of-freedom (3-DOF) gait simulator for evaluating transtibial prostheses. The proposed platform integrates custom mechanical, electrical, electronic, and embedded software components to reproduce the horizontal, vertical, and sagittal motions of the human gait cycle.

The functional assessment demonstrated low tracking error in the reproduction of the tibial inclination angle during both the stance and swing phases. In addition, a strong waveform correlation with the reference measurements was achieved for the vertical ground reaction force during the stance phase. However, differences in peak magnitude were observed and attributed to factors not characterized in this study, such as the added mass of the simulator's moving assembly and the mechanical properties of the commercial prosthesis.

The study demonstrates the preliminary feasibility of a repeatable, position-driven 3-DOF gait simulator, assessed with a single participant. Further independent validation of all motion axes, improved kinetic tracking, structural durability testing, and experiments comparing different prosthesis configurations are required before the platform can be considered validated for transtibial prosthesis evaluation. Future work will also extend the assessment to multiple subjects with varying anthropometric characteristics, gait patterns associated with foot pathologies, and prototype powered transtibial prostheses equipped with closed-loop control.

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